Subject: Mathematics **WEEKLY CHAPTER TEST – 04 (PROBABILITY)** **Time:** 45 Minutes
Class: Class X (CBSE) **Chapter 14: Probability** **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Section A has 4 MCQs and 1 Assertion-Reason question (1 mark each).
2. Section B has 1 VSAQ question (2 marks). Section C has 3 SA questions (3 marks each).
3. Section D has 1 Applied Case Study (4 marks). Show all working clearly.

SECTION A: OBJECTIVE & MCQs (1 Mark Each) [5 Marks]

1. Two dice are rolled together. The probability of obtaining a total score of 7 is: **[1]**

Options: (a) $\frac{1}{6}$ (b) $\frac{7}{36}$ (c) $\frac{5}{36}$ (d) $\frac{1}{12}$

2. All the red face cards are removed from a well-shuffled pack of 52 cards. A card is then drawn at random from the remaining cards. The probability that the drawn card is a red card is: **[1]**

Options: (a) $\frac{20}{46}$ (b) $\frac{10}{23}$ (c) $\frac{1}{2}$ (d) $\frac{13}{23}$

3. A box contains 90 discs which are numbered from 1 to 90. If one disc is drawn at random from the box, the probability that it bears a two-digit number is: **[1]**

Options: (a) $\frac{9}{10}$ (b) $\frac{89}{90}$ (c) $\frac{81}{90}$ (d) $\frac{4}{5}$

4. A box contains 400 apples. The probability of getting a rotten apple in the box is 0.035. The total number of rotten apples in the box is: **[1]**

Options: (a) 7 (b) 14 (c) 21 (d) 28

5. **Assertion (A):** The probability of drawing a face card from a well-shuffled pack of 52 cards is $\frac{3}{13}$.
Reason (R): In a deck of 52 cards, there are 12 face cards (4 Kings, 4 Queens, 4 Jacks). **[1]**

(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

SECTION B: VERY SHORT ANSWER QUESTIONS (2 Marks Each) [2 Marks]

6. A bag contains 5 red, 8 green, and 7 white balls. One ball is drawn at random from the bag. What is the probability that the ball drawn is: (i) white (ii) not green? **[2]**

SECTION C: SHORT ANSWER QUESTIONS (3 Marks Each) [9 Marks]

7. Three unbiased coins are tossed together. Find the probability of getting: (i) all heads (ii) at least two heads (iii) at most two heads. **[3]**

8. The king, queen, and jack of clubs are removed from a deck of 52 playing cards and the remaining cards are well shuffled. A card is drawn at random from the remaining cards. Find the probability of getting: (i) a club (ii) a face card (iii) a red card. **[3]**

9. A box contains 90 discs numbered from 1 to 90. If one disc is drawn at random from the box, find the probability that it bears: (i) a two-digit number (ii) a perfect square number (iii) a number divisible by 5. **[3]**

SECTION D: APPLIED CASE STUDY (4 Marks) [4 Marks]

Question 10: Case Study Analysis

"Eight Ball" is a popular cue sport played on a pool table with 15 object balls numbered 1 to 15, and one white cue ball. The object balls are partitioned as follows:

- Balls 1 to 7 are solid colored.
- Ball 8 is solid black.
- Balls 9 to 15 are striped. On a break shot, one object ball is pocketed at random.

Based on the above scenario, answer the following:

(i) What is the probability that the pocketed ball is a solid colored ball (from 1 to 7)? **[1]**

(ii) What is the probability that the pocketed ball is an even-numbered striped ball? **[1]**

(iii) What is the probability that the pocketed ball has a prime number? **[1]**

(iv) What is the probability that the pocketed ball is the 8-ball? **[1]**

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